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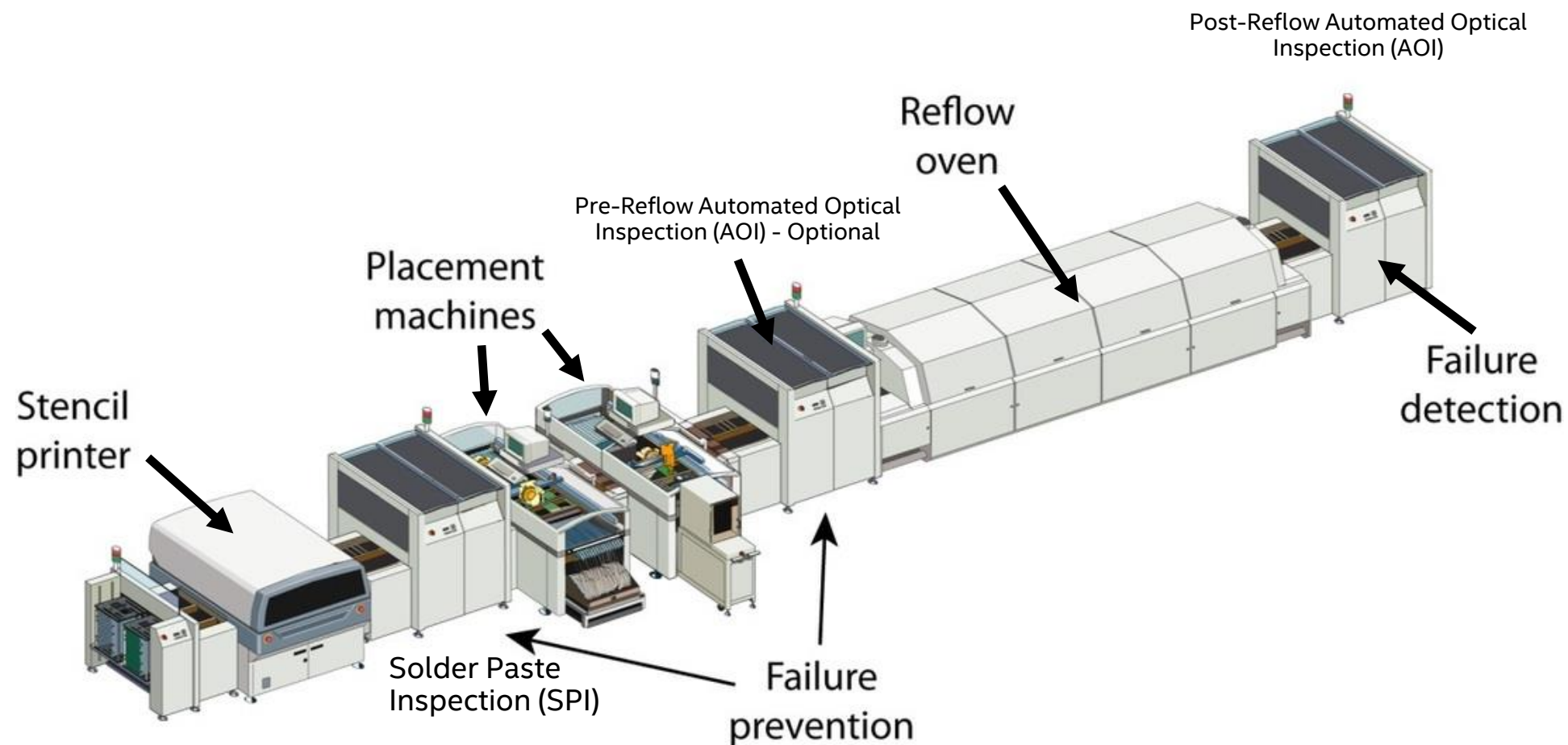
2020 ADL Customers CRD Summary Report

Rev1.2, CQ&R CMEE Team, 2020Q4



intel[®]

General SMT Line Layout



Objectives

Share Findings From CMEE Mobile Industry Data Collection (ADL Platform Customer Survey)

- Data Collected From 10 ODM Customers/Sites
 - Compal (KunSan – Build for Dell/HP, ChengDu – Build for Dell, ChongQing – Build for Acer)
 - HuaQin (Build for HuaWei/Asus/Lenovo)
 - Inventec ChongQing (Build for HP)
 - LCFC (Build for Lenovo)
 - MSI KunSan (Own Brand)
 - Pegatron (ChongQing – Build for Asus, SuZhou – Build for MSFT/Intel)
 - Wistron TaiWan (Build for Lenovo)

Highlights

- See increasing usage of Yunnan-Tin and other pastes, ShenMao is reducing the market share in client segment ➔
- More customers capable on 01005 component SMT ➔
- 0.35mm pitch BGA acceptance still relatively low (3/10 capable) ➔
- See increased customer adoption of vacuum support block ➔
- Copper-Nickel-Zinc is a quite popular material for EMI shield ➔
- Black SM PCB downgrade AOI performance, SMD pad preferred for small size Pad ➔
- All customers measure O2 PPM at reflow/peak zone ➔
- UV glue is the most common board adhesive, apply after SMT before test ➔
- 2mm is the most common KOZ width, different customers have KOZ length ➔
- There are 3 customers are using N2 on rework process ➔
- Black SM cost up 3%~5%, Tg 150 is the most common material ➔

Solder Paste Printing



Solder Paste Printing (1 Of 5)

What are the top 3 SAC pastes and the percentage of each type used in your production line?				
Compal KS	Compal CD	Compal CQ	HuaQin	Inventec CQ
YuanNan Tin YW9-3005-98CY04 (96%) ShenMao PF606-P (2%) YankTai NP01-P101-I+ (2%)	YuanNan Tin YW9-3005-98CY04 (100%) ShenMao PF606-P (0%) * Percentage could be change overtime	YuanNan Tin YW9-3005-98CY04 (90%) ShenMao PF606-P (10%)	Shenmao PF606-P26 SAC305 Type4: 87.5% Shenmao PF606-P245 SAC305 Type4: 9.37% INDIUM8.9HF SAC305 Type5-MC: 3.13%	Shenmao PF606-P (50%) TamuraTLF204 (40%) Eunow EUP1000 HQR (T5)(10%)
LCFC	MSI	Pega CQ	Pega SZ (JDM1 + Intel SPG & NSG)	Wistron
Shenmao PF606 (70%) * Other 30% is LTS paste	KOKI M406-3	ShenMao PF606-P YunNan Tin YW9-3005-98US04, Type-4, SAC305 * US04: YunNan Tin customized for Pegatron/Asus only	Intel NUC: DW-1000, SSD: PF606-P25 1.Eunow LTS DW-1000 2.Shenmao PF606-P26&PF606-P 3. Senju S70G&M705-S101HF(N4) 4.Indium 8.9HFA Type 4.5	SAC305, SAC105

Seeing Trend Of Reduction Of ShenMao And Increasing Of YunNan Tin



Solder Paste Printing (2 Of 5)

What is the Stencil Thickness Range (mm) used for different package pitch?				
Compal KS	Compal CD	Compal CQ	HuaQin	Inventec CQ
0.4mm: 3mil 0.5~0.6mm: 4~5mil	0.4~0.5mm: 4mil	0.65mm: 5mil	4mil – 5mil	0.35mm: 3mil others: 4~5mil
LCFC	MSI	Pega CQ	Pega SZ (JDM1 + Intel SPG & NSG)	Wistron
0.35~0.5mm :3mil >=0.65mm: 4mil or 5mil	0.5~0.6mm: 4mil 0.7mm: 5mil	0.4mm: 3mil 0.6mm: 4mil	0.35mm-3mil 0.4mm-3mil/4mil 0.5mm-4mil 0.6mm-5mil	3mil ~ 5mil
If 01005 component size is used on ADL-M platform, is your mfg. line qualified for HVM assembly?				
Compal KS	Compal CD	Compal CQ	HuaQin	Inventec CQ
Yes	No	No	Yes, already in HVM	Yes
LCFC	MSI	Pega CQ	Pega SZ (JDM1 + Intel SPG&NSG)	Wistron
Yes, so far less than 20Pcs per panel	No	No	Machine capable, no real HVM yet	Yes

3 Mil For 0.35~0.4 Pitch BGA, 4~5 Mil For 0.5~0.7mm Pitch BGA
6/10 Customers Have 01005 Capability

Solder Paste Printing (3 Of 5)

Do you require Intel to provide 3mil a stencil recommendation for U-Series CPU? If yes, can you specify the reason?				
Compal KS	Compal CD	Compal CQ	HuaQin	Inventec CQ
No	Nice to have, future for small pitch connector we may use 3mil stencil	No	No	No
LCFC	MSI	Pega CQ	Pega SZ (JDM1 + Intel SPG & NSG)	Wistron
No, 0.65mm pitch keep use same 4~5mil stencil	No requirement, but nice to have	Nice to have, for fine pitch connector	No	Depends on customer requirement
What is smallest pad to pad separation for 1 mil (25um) step down stencil?				
Compal KS	Compal CD	Compal CQ	HuaQin	Inventec CQ
1mm	0.4mm	NA	0.5mm	7.5mm
LCFC	MSI	Pega CQ	Pega SZ (JDM1 + Intel SPG&NSG)	Wistron
0.2mm	5mm	5mm	5mm	Confidential

3mil For U Package Is Not A Major Request Yet
Very Wide Range Of Pad To Pad Gap For Step Stencil



Solder Paste Printing (4 Of 5)

What is the smallest AR (Area Ratio) capable in HVM?				
Compal KS	Compal CD	Compal CQ	HuaQin	Inventec CQ
0.6	0.68	0.66	0.66	0.55
LCFC	MSI	Pega CQ	Pega SZ (JDM1 + Intel SPG & NSG)	Wistron
0.66	0.66	0.65	0.66	Confidential
Do you have challenges printing 0.35mm pitch BGAs?				
Compal KS	Compal CD	Compal CQ	HuaQin	Inventec CQ
Yes	Yes	Yes	No	Yes
LCFC	MSI	Pega CQ	Pega SZ (JDM1 + Intel SPG&NSG)	Wistron
Yes, need to use whole process pallet	Yes	Yes	No (in using)	No

Most Customers Still Hold 0.66 AR
3/10 Customers Think 0.35 Pitch BGA Is Not A Challenge



Solder Paste Printing (5 Of 5)

Pictures

For Intel SOC board side, and following pcb thicknesses of 0.6, 0.7, 0.8 & 1.0 mm, please state if support pins, print pallets, vacuum fixture, or support block is used for board support.				
Compal KS	Compal CD	Compal CQ	HuaQin	Inventec CQ
Vacuum Block	Support block	Support Block	≤0.8mm: print pallets; >0.8mm: support block	Support Block
LCFC	MSI	Pega CQ	Pega SZ (JDM1 + Intel SPG & NSG)	Wistron
For all thickness, first side use support block, 2nd side use vacuum block	Support Pin	<0.8mm: Whole process pallet >=0.8mm: Support block	Whole process pallet + Support block	Depends
Do you use different board support strategies for different board thicknesses: 0.6, 0.7, 0.8, 1.0 mm?				
Compal KS	Compal CD	Compal CQ	HuaQin	Inventec CQ
No	No, all brd are 1.0mm in our site	No	≤0.8mm: print pallets; >0.8mm: support block	No
LCFC	MSI	Pega CQ	Pega SZ (JDM1 + Intel SPG&NSG)	Wistron
No	All use support Pin	<0.8mm: print pallets; >=0.8mm : support block	All board SOC side use whole process pallet + Support block	Depends

Support Block Still The Mainstream. Vacuum Block Start Appear In Customer Line. Print Pallet Will Be Used For Thin Brd.

Solder Paste Inspection (SPI)

Solder Paste Printing (1 Of 1)

For SPI volume control limits, what is percentage of defects (opens, in-sufficients, bridging) attributed to print volume?				
Compal KS	Compal CD	Compal CQ	HuaQin	Inventec CQ
0.03%	No answer	No answer	0%	0.3% out of all the failures
LCFC	MSI	Pega CQ	Pega SZ (JDM1 + Intel SPG & NSG)	Wistron
0.08%	near to 0%	No answer	No data	Confidential
What is the typical percentage rate of board wash?				
Compal KS	Compal CD	Compal CQ	HuaQin	Inventec CQ
1/2000	4/4500	2/1500	0.11%	1/1500
LCFC	MSI	Pega CQ	Pega SZ (JDM1 + Intel SPG&NSG)	Wistron
0.25%	near to 0%	near to 0%	Avg 4/1000	Confidential

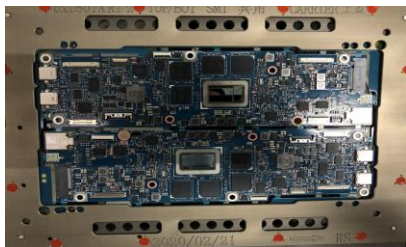
Pick and Placement (PnP) Machine

Pick and Placement (PnP) Machine (1 of 2)

State BGA Placement Force in Newtons when Intel component is placed?				
Compal KS	Compal CD	Compal CQ	HuaQin	Inventec CQ
2.5N	2.5N	2.5N	N/A	N/A
LCFC	MSI	Pega CQ	Pega SZ (JDM1 + Intel SPG & NSG)	Wistron
2.5N	N/A	0	NXT: 0 NPM:2.5N	Confidential
For Intel SOC board side, and following pcb thicknesses of 0.6, 0.7, 0.8 & 1.0 mm, please state if placement pallets are used for board support.				
Compal KS	Compal CD	Compal CQ	HuaQin	Inventec CQ
No	No	No	No	No (except when use whole proces pallet)
LCFC	MSI	Pega CQ	Pega SZ (JDM1 + Intel SPG&NSG)	Wistron
No	No	No (except when use whole proces pallet)	Whole process pallet	Whole process pallet

Nobody Is Using PnP Pallets Expect When Use Whole Process Pallets

Pick and Placement (PnP) Machine (2 of 2)

Do you use different board support strategies in PnP machine for different board thicknesses: 0.6, 0.7, 0.8, 1.0 mm?				
Compal KS	Compal CD	Compal CQ	HuaQin	Inventec CQ
No	Maybe	No answer	<0.8mm, use whole process pallet	PCB = 0.6mm, use whole process pallet PCB > 0.6mm, no pallet
LCFC	MSI	Pega CQ	Pega SZ (JDM1 + Intel SPG & NSG)	Wistron
No	No	No (except when use whole process pallet)	Whole process pallet	Depends on customer requirement
Please provide images of your different board support types.				
Compal KS	Compal CD	Compal CQ	HuaQin	Inventec CQ
N/A	No answer	No answer	No answer	
LCFC	MSI	Pega CQ	Pega SZ (JDM1 + Intel SPG&NSG)	Wistron
N/A	N/A			No answer

Large EMI Shields (> 40 x 40 mm)

Large EMI Shields (> 40 x 40 mm) (1 of 3)

What is the preferred method for implementing large EMI Shielding (> 40 x 40 mm) in your designs?				
Compal KS	Compal CD	Compal CQ	HuaQin	Inventec CQ
A and B	A	C	B	B
LCFC	MSI	Pega CQ	Pega SZ (JDM1 + Intel SPG & NSG)	Wistron
A and B	C	B	A	Depends on customer requirement
If answered "Option A" to the question above: Do you utilize hold down top hat/weight on top of large EMI shields (> 40 x 40 mm) during reflow to reduce risk of open solder joint?				
Compal KS	Compal CD	Compal CQ	HuaQin	Inventec CQ
Yes	No, so far we don't have such big shielding cap	NA	No answer	N/A
LCFC	MSI	Pega CQ	Pega SZ (JDM1 + Intel SPG&NSG)	Wistron
No	N/A	N/A	Yes	No answer

Option A: 4/9; Option B: 5/9; Option C: 2/9
Option A: 2/4 Using Top Hat

Option A: Soldered Down directly to PCB in SMT Process
Option B: SMT Clips soldered down in SMT, then LID added post SMT Assy
Option C: No shields added in SMT, we utilize System for shielding our designs

Large EMI Shields (> 40 x 40 mm) (2 of 3)

What is the typical size EMI Shield pad width (example 0.5 mm or larger) you allow for option A or B, for large Soldered-down EMI shields (> 40 x 40 mm)?				
Compal KS	Compal CD	Compal CQ	HuaQin	Inventec CQ
0.8mm	No answer	N/A	0.5mm&1.0mm	No answer
LCFC	MSI	Pega CQ	Pega SZ (JDM1 + Intel SPG & NSG)	Wistron
0.6mm	N/A	>= 0.5mm	0.5mm	Confidential
Do you overprint solder paste for these large EMI Soldered down shields (> 40 x 40 mm)? If yes, what is typical overprint % to pad size?				
Compal KS	Compal CD	Compal CQ	HuaQin	Inventec CQ
Yes, <170%	>200%	N/A	0.5mm→1.0mm overprint 100%	No answer
LCFC	MSI	Pega CQ	Pega SZ (JDM1 + Intel SPG&NSG)	Wistron
1:1 (no overprint)	N/A	100-300%	250-300%	Confidential
What geometry (rectangular, circle, oval, other) do you use for shield pads on large EMI soldered down shields (> 40 x 40 mm)?				
Compal KS	Compal CD	Compal CQ	HuaQin	Inventec CQ
Rectangular	Rectangular	N/A	Rectangular	Rectangular
LCFC	MSI	Pega CQ	Pega SZ (JDM1 + Intel SPG & NSG)	Wistron
Rectangular	N/A	Rectangular	Rectangular	No answer



Large EMI Shields (> 40 x 40 mm) (3 of 3)

What material is used for shield part - lid and fence for large EMI soldered down shields (> 40 x 40 mm)?				
Compal KS	Compal CD	Compal CQ	HuaQin	Inventec CQ
4: Copper-Nickel-Zinc Alloy	No answer	N/A	2, 4 (Copper-Nickel-Zinc Alloy)	No answer
LCFC	MSI	Pega CQ	Pega SZ (JDM1 + Intel SPG & NSG)	Wistron
4: Nickel Sliver Copper (1?)	N/A	No answer	2, 4 (Copper-Nickel-Zinc)	No answer
Do you utilize any solder additive process for the shield reflow to increase mechanical stability, for large EMI soldered down shields (> 40 x 40 mm)?				
Compal KS	Compal CD	Compal CQ	HuaQin	Inventec CQ
No	No answer	N/A	No	small solder chips that can be placed by SMT machine
LCFC	MSI	Pega CQ	Pega SZ (JDM1 + Intel SPG&NSG)	Wistron
No	N/A	No	No	No answer

Copper-Nickel-Zinc Is A Quite Popular Material

Typical Fence Materials:

- 1) NiAg
- 2) Tin Plated CRS (Cold Rolled Steel)
- 3) 1000 Series Al
- 4) Other?

Reflow Convection Oven



Reflow Convection Oven (1 of 3)

Do you see any technical issue in SMT process if using Black Color Soldermask PCB?				
Compal KS	Compal CD	Compal CQ	HuaQin	Inventec CQ
No	We have no black solder mask PCB	We have no black solder mask PCB	Black Color Soldermask will interfere with AOI image recognition components	We have no black solder mask PCB
LCFC	MSI	Pega CQ	Pega SZ (JDM1 + Intel SPG & NSG)	Wistron
No	No	Black Color Soldermask will interfere with AOI image recognition components	Yes 1. Black color soldermask will interfere with AOI to recognition small resistors (black color), cannot catch resistor missing 2. Not easy to keep good quality of small size MD pad	No

- 3/10 Customers Have No Black SM Experience
- Major Problem Of Black SM:
 1. AOI interference causing downgrade of detect capability.
 2. Can not achieve good quality of small size MD pad. SMD pad preferred



Reflow Convection Oven (2 of 3)

If N2 is used, at what zone numbers is the O2 PPM level measured?				
Compal KS	Compal CD	Compal CQ	HuaQin	Inventec CQ
Zone 9	Zone 9~10	No answer	Zone 10 (total 13 zones)	Peak Zone
LCFC	MSI	Pega CQ	Pega SZ (JDM1 + Intel SPG & NSG)	Wistron
Peak Zone	Peak Zone	No build-in O2 analyzer. We use standalone equipment	Peak Zone	Peak Zone
Do thinner boards require different support vs thicker? (example: please state answer for pcb thickness 1.0mm, 0.8mm, 0.7mm, 0.6 mm)				
Compal KS	Compal CD	Compal CQ	HuaQin	Inventec CQ
No	No, all brd are 0.1mm in our site	No	Thinner PCB will use more support pins	No
LCFC	MSI	Pega CQ	Pega SZ (JDM1 + Intel SPG&NSG)	Wistron
No, all PCB 0.8mm~1.6mm now	No, All use reflow pallet	<0.8mm: Whole process pallet >=0.8mm: Reflow only pallet	No, All use dedicate pallet	Yes

Customer Are Measuring O2 PPM At Peak Zone
No Difference Pallet Support Unless Use Whole Process Pallet

Reflow Convection Oven (3 of 3)

How are PCBs placed into reflow pallets? Manually, robotic?				
Compal KS	Compal CD	Compal CQ	HuaQin	Inventec CQ
Robotic	Robotic	Robotic	Robotic	Robotic
LCFC	MSI	Pega CQ	Pega SZ (JDM1 + Intel SPG & NSG)	Wistron
Both	Manual	Robotic	Whole process pallet	Manual
Is the pallet common to print through reflow or reflow only?				
Compal KS	Compal CD	Compal CQ	HuaQin	Inventec CQ
Reflow only	Reflow only	Reflow only	Reflow only	Reflow only
LCFC	MSI	Pega CQ	Pega SZ (JDM1 + Intel SPG&NSG)	Wistron
Whole process pallet for PCB w/o outrigger. Dedicate reflow pallet for normal PCB	Reflow only	<0.8mm: Whole process pallet >=0.8mm: Reflow only pallet	Whole process pallet	Depends on product
If tooling Pins on pallet are used for pcb hold down, what is the clearance in design (mm) used to allow for board expansion without binding during reflow cycle?				
Compal KS	Compal CD	Compal CQ	HuaQin	Inventec CQ
1mm	2mm	0.5mm	1mm	0.5mm
LCFC	MSI	Pega CQ	Pega SZ (JDM1 + Intel SPG & NSG)	Wistron
1mm	There is no cavity on the reflow pallet	1mm	1mm	No answer

Adhesive SJR (Solder Joint Reliability)

Adhesive SJR (1 of 12)

Are you currently using adhesive for Intel BGA in HVM?				
Compal KS	Compal CD	Compal CQ	HuaQin	Inventec CQ
Yes	Yes	Yes	Yes	Yes
LCFC	MSI	Pega CQ	Pega SZ (JDM1 + Intel SPG & NSG)	Wistron
Yes	Yes	Yes for CPU/GPU/PCH	Yes, some projects	Depends on product

**All Customers Are Using Adhesive On Intel BGAs, 8/10
Customers Will 100% Apply, 2/10 Customers Will Depends On
Products**

Adhesive SJR (2 of 12)

Please state which adhesive process you are using										
	Compal KS	Compal CD	Compal CQ	HuaQin	Inventec CQ	LCFC	MSI	Pega CQ	Pega SZ (JDM1 + Intel SPG & NSG)	Wistron
Corner Glue	Yes	Yes	Yes	Yes	Yes (Thermal)		Yes	Yes	Yes (Thermal)	Yes (Unknown)
Corner Dot										Yes
Corner Partial underfill				Yes		Yes			Yes	Yes
Full Underfill;									Yes	Yes
No Flow Underfill										
Underfilm										
SJEM Epoxy Flux										Have the capability, but not in use



Adhesive SJR (3 of 12)

CG* - Corner Glue
 PU* - Partial Underfill
 CD* - Corner Dot
 BLUF* - Board Level Underfill
 PU¹ - JDM1 experienced SJ crack after test, the solution is move partial underfill to before test

What's the vendor name and model of the adhesive material you are using for each of the following processes (Please provide Vendor Name & Model for each)?				
Compal KS	Compal CD	Compal CQ	HuaQin	Inventec CQ
Loctite 3705(UV); SOMIDA(索梦达) 3805 (UV)	No answer	Ubond UB-4305 (UV)	Loctite 3705 (UV) Syscotech(新懿) SCT 3109A (Corner Underfill)	Loctite 3619(Thermal)
LCFC	MSI	Pega CQ	Pega SZ (JDM1 + Intel SPG & NSG)	Wistron
Syscotech(新懿) SCTAS3109A/AS3109A7 (Corner Underfill)	ZhongYu ZY-330 (UV)	SeaYu(信友) UV3329 (UV)	SeaYu(信友) UV3329 (UV) Zymet UA2605-B (Thermal)	Confidential
Where in the board manufacturing process is the adhesive applied (For each, please specify "Before SMT", "After SMT/Before Functional Test", or "After SMT/After Functional Test".				
Compal KS	Compal CD	Compal CQ	HuaQin	Inventec CQ
After SMT before test (CG*)	After SMT before test (CG*)	After SMT before test (CG*)	After SMT before test (CG*) After function test (PU*)	After SMT before test (CG*)
LCFC	MSI	Pega CQ	Pega SZ (JDM1 + Intel SPG&NSG)	Wistron
After function test (PU*)	After SMT before test (CG*)	After SMT before test (CG*)	After SMT, before test (CG*, PU ¹)	After SMT, before test (CG*, PU*) Before SMT (CD*) Have both before or after test (BLUF*)

UV Glue Is The Major Adhesive Most Customers Are Using, With Many Different Brand. Seeing Trend Of Moving Adhesive Process To 'After SMT Before Test' Even With BLUF* And PU* process

Adhesive SJR (4 of 12)

If using Corner Glue, what % of the Perimeter length of Package is covered?				
Compal KS	Compal CD	Compal CQ	HuaQin	Inventec CQ
15~20%	5mm L shape	Follow DFX silkscreen	26%	4mm L shape
LCFC	MSI	Pega CQ	Pega SZ (JDM1 + Intel SPG & NSG)	Wistron
N/A	L shape cover 4~6 row/columns each side	Follow DFX silkscreen	30%~50%	Depends on customer requirement
If using Corner Glue, what is minimum % of substrate fillet height acceptable?				
Compal KS	Compal CD	Compal CQ	HuaQin	Inventec CQ
60%~100%	<100%	80%	90%-100%	50%
LCFC	MSI	Pega CQ	Pega SZ (JDM1 + Intel SPG&NSG)	Wistron
N/A	No answer	<100%	75%	Depends on customer requirement

Adhesive SJR (5 of 12)

If using Corner Dot, how many dots are placed per package? State where used (Corners, mid-point of sides, edges)?

Wistron

Depends on customer requirement

If using Corner Partial Underfill, what is the minimum & maximum rows or columns of coverage?

HuaQin	LCFC	Pega SZ (JDM1)	Wistron
At least 8 columns/rows per corner (Corner fill)	At least 8 columns/rows per corner (Corner fill) SAC305:twice jet at corner LTS: twice jet at corner and edge 	3x3	Depends on customer requirement

If using Underfill what is the maximum cumulative voiding % of dispensed area acceptable (<10%, >10%<25%, >25%, <50%): IPC J-STD-030 <10% acceptable?

HuaQin	LCFC	Pega SZ (JDM1)	Wistron
<10%	No Answer	Follow IPC	Depends on customer requirement

If using Underfill, what is minimum % of substrate fillet height acceptable?

HuaQin	LCFC	Pega SZ (JDM1)	Wistron
No answer	No answer	50%	Depends on customer requirement

If using Underfill what are methods for inspection to quantify voiding? (cross-section, C-SAM, Dye and Pry, Other)

HuaQin	LCFC	Pega SZ (JDM1)	Wistron
No answer	No answer	X-section	X-section

Adhesive SJR (6 of 12)

If using Underfill, Is adhesive encapsulation acceptable on adjacent caps/resistor/inductors around the perimeter of the BGA packages?			
HuaQin	LCFC	Pega SZ (JDM1)	Wistron
No	No	Yes	Depends on customer requirement
If using Underfill, is it acceptable for adhesive material to flow on land side capacitors/passives on underside of BGA package? (Please answer Yes/No)			
HuaQin	LCFC	Pega SZ (JDM1)	Wistron
Yes	No Answer	Yes	Depends on customer requirement
If using Underfill, are voids acceptable? (Please answer Yes/No)			
HuaQin	LCFC	Pega SZ (JDM1)	Wistron
Yes	No answer	No	Depends on customer requirement
If using a material that flows underneath the package, is partial encapsulation of LSC's (Land Side Capacitors) acceptable?			
HuaQin	LCFC	Pega SZ (JDM1)	Wistron
Yes	Yes	Yes	Depends on customer requirement

Adhesive SJR (7 of 12)

If using No Flow Underfill what is the maximum cumulative voiding % of dispensed area criteria?

If using No Flow Underfill what is the minimum % of substrate fillet height acceptable ?

If using Underfilm, what is the minimum % coverage of perimeter length around package?

If using SJEM Epoxy Flux (Solder Joint Encapsulation Material), what is the minimum acceptable Solder Joint encapsulation % with epoxy material, as measured post reflow by cross-section or other means?

No Answers For Those Questions Because Customers Who Response The Survey Are Not Using Those Adhesive Materials

Adhesive SJR (8 of 12)

Is there a preference of 1 Part Epoxy or 2 Part Epoxy in material selection/usage?				
Compal KS	Compal CD	Compal CQ	HuaQin	Inventec CQ
No answer	No answer	No answer	No	No
LCFC	MSI	Pega CQ	Pega SZ (JDM1 + Intel SPG & NSG)	Wistron
No	No	No	No	No
What is the minimum acceptable Epoxy Pot Life use condition (in hours)?				
Compal KS	Compal CD	Compal CQ	HuaQin	Inventec CQ
UV: 2 weeks if no expose to light	No answer	72h	24h	No answer
LCFC	MSI	Pega CQ	Pega SZ (JDM1 + Intel SPG&NSG)	Wistron
No answer	Follow datasheet	No answer	24h	No answer

Adhesive SJR (9 of 12)

What type of dispensing valve is used for Automated Dispense? (For example, Pneumatic, Piezo, Auger, Other)				
Compal KS	Compal CD	Compal CQ	HuaQin	Inventec CQ
Pneumatic	Pneumatic, Piezo	Pneumatic	Pneumatic	Auger
LCFC	MSI	Pega CQ	Pega SZ (JDM1 + Intel SPG & NSG)	Wistron
Auger	Piezo	Pneumatic	Pneumatic	Piezo
Is the machine able to heat the nozzle, and if YES, what is the max temperature (Celsius) the machine is capable?				
Compal KS	Compal CD	Compal CQ	HuaQin	Inventec CQ
No	No	No	No	No
LCFC	MSI	Pega CQ	Pega SZ (JDM1 + Intel SPG&NSG)	Wistron
Yes, 50C	No	No	No	Yes, 100C
Is there ability to heat the substrate of the board, and if YES, what is the max bottom heater temperature (Celsius) the machine is capable?				
Compal KS	Compal CD	Compal CQ	HuaQin	Inventec CQ
No	No	No	No	No
LCFC	MSI	Pega CQ	Pega SZ (JDM1 + Intel SPG & NSG)	Wistron
Machine support the function; The highest temp can reach to 210C. But we didn't buy the module.	No	No	No	Yes, 130C

Adhesive SJR (10 of 12)

What is the Dispense Machine Info (Vendor Name, Model Number, Model ID, Model Year)?				
Compal KS	Compal CD	Compal CQ	HuaQin	Inventec CQ
No answer	No answer	No answer	NORDSON, Q-6800, Q30380, 2019	No answer
LCFC	MSI	Pega CQ	Pega SZ (JDM1 + Intel SPG & NSG)	Wistron
AXXON/AI100S/01DPGAXX0130021/2019-09-03	AXXON	No answer	ANDA, HD-18SHW, Year 2015	Asymtek
For In-line thermal Cure Oven, Batch thermal Cure Oven, In-line UV Cure Oven being used, please state Vendor name, Model #, Model ID & Model Year.				
Compal KS	Compal CD	Compal CQ	HuaQin	Inventec CQ
Elite Automatic, UV703, 2018/09	No answer	No answer	KTX-600, 19LSL01, 2019	No answer
LCFC	MSI	Pega CQ	Pega SZ (JDM1 + Intel SPG&NSG)	Wistron
Dongsheng/BTA-1500-VF/01CURDON0130003/2012-10-29	Elite	億立/宏創/台台	Heller 1900 Year: 2003 SN# 0203F2A7N5-34742-01	Rehm

Adhesive SJR (11 of 12)

Is there any flux compatibility testing performed? (flux contained in solder paste with adhesive material used)

If Yes to the above question, what test/equipment is used? (For example, DSC, Other)

No Customers Had Performed Flux Compatibility Testing



Adhesive SJR (12 of 12)

What adhesive keep out zone (KOZ) in mm is required for adhesive dispense?				
Compal KS	Compal CD	Compal CQ	HuaQin	Inventec CQ
2mm	2mm*5mm	Follow DFX silkscreen	2mm	2mm
LCFC	MSI	Pega CQ	Pega SZ (JDM1 + Intel SPG & NSG)	Wistron
Min:1mm, jet apply	2mm	0.5mm	2mm	Depends on customer requirement
How is after cure validated, for thermal cure adhesive material, non-destructively (voiding, area coverage, adhesion to pcb and component)				
Compal KS	Compal CD	Compal CQ	HuaQin	Inventec CQ
N/A	N/A	N/A	Visual inspection	The adhesion between PCB and component
LCFC	MSI	Pega CQ	Pega SZ (JDM1 + Intel SPG&NSG)	Wistron
Use rework method to remove the BGA then do visual inspection	N/A	Visual inspection	verify the gluing process by mount the component on glass plate	N/A

2mm Is The Most Common KOZ Width

Rework



Rework (1 of 2)

Is N2 used for rework? If yes, what is the O2 PPM level?				
Compal KS	Compal CD	Compal CQ	HuaQin	Inventec CQ
No	No	No	No	No
LCFC	MSI	Pega CQ	Pega SZ (JDM1 + Intel SPG & NSG)	Wistron
No	Yes, under 1000dppm	No	Yes, not measuring the O2 PPM	Yes
Do you see any technical issue in Rework process if using Black Color soldermask PCB?				
Compal KS	Compal CD	Compal CQ	HuaQin	Inventec CQ
No	N/A	No experience	Pay attention to 0.4pitch BGA, black SM normally use SMD pad. Need x-ray double confirm after rework	No
LCFC	MSI	Pega CQ	Pega SZ (JDM1 + Intel SPG&NSG)	Wistron
No	No	No	No	No

**A Few Customers Start Use N2 For Rework
Black SM Doesn't Impact Rework Process**

Rework (2 of 2)

How do you decide use paste or Flux used to rework Intel SOC?				
Compal KS	Compal CD	Compal CQ	HuaQin	Inventec CQ
Depends on Pitch	Use paste	Use paste	Use paste	Use paste
LCFC	MSI	Pega CQ	Pega SZ (JDM1 + Intel SPG & NSG)	Wistron
Depends on BGA pitch	Use paste	Pitch > 0.3mm: Paste Pitch <=0.3mm: Flux	All BGA use paste	Depends on customer requirement
Do you have any challenges reworking 0.35 mm pitch BGA (site re-dress, flux dispense, placement)?				
Compal KS	Compal CD	Compal CQ	HuaQin	Inventec CQ
Yes	Yes	Yes. 1. Need new rework machine; 2, Easy damage SM when do pad clean; 3. Need auto placement for accurate alignment	No	Yes, hard to print paste and keep good alignment
LCFC	MSI	Pega CQ	Pega SZ (JDM1 + Intel SPG&NSG)	Wistron
No	No 0.35mm pitch BGA now	Yes, prefer >=0.4mm	Yes, difficult for paste alignment causing SBB	No

A Lot Of Customers Use Paste Only Rework
Most Customers Still Have Challenges On 0.35mm BGA Rework

Bare PCB (Printed Circuit Board)

Bare PCB (1 of 3)

State PCB Thickness Range for Type 3 (Through Hole)				
Compal KS	Compal CD	Compal CQ	HuaQin	Inventec CQ
0.8~1.2mm	1.0mm	1.0mm	0.8mm, 1.0mm	0.8~1.2mm
LCFC	MSI	Pega CQ	Pega SZ (JDM1 + Intel SPG & NSG)	Wistron
0.9-1.0mm	1.0mm, 1.2mm	0.8-1.3mm	0.8mm~0.85mm, 0.9mm	No answer
State PCB Thickness Range for Type 4 (HDI) High Density Interconnect				
Compal KS	Compal CD	Compal CQ	HuaQin	Inventec CQ
0.7~1.2mm	N/A	N/A	1.0mm	0.6~1.2mm
LCFC	MSI	Pega CQ	Pega SZ (JDM1 + Intel SPG&NSG)	Wistron
0.8-1.2mm	N/A	No answer	0.8mm~0.85mm, 0.9mm	No answer

0.8mm~1.2mm Is The Most Common Thickness

Bare PCB (2 of 3)

State Top 3 PCB Vendors Type 3 by volume				
Compal KS	Compal CD	Compal CQ	HuaQin	Inventec CQ
GCE; Tripod; Hannstar	TMT,GCE,Tripod,Hannstar	chuanyi 40%/TMT 60%	ASKPCB 30%; HSB(Hannstar)30%; CEEPCB 20%; ShenHong Tech 20% (http://www.shpcb.com/)	No answer
LCFC	MSI	Pega CQ	Pega SZ (JDM1 + Intel SPG & NSG)	Wistron
No answer/Confidential	Hanstar,Tripod	No answer/Confidential	UMT, Compeq, Tripod, Hanstar, GCE	No answer
State Top 3 PCB Vendors Type 4 by volume				
Compal KS	Compal CD	Compal CQ	HuaQin	Inventec CQ
GCE; Tripod; Hannstar	N/A	N/A	Avary Holding 60%(http://www.avaryholding.com/) ; Unimicron 40%	No answer
LCFC	MSI	Pega CQ	Pega SZ (JDM1 + Intel SPG&NSG)	Wistron
No answer/Confidential	No answer/Confidential	No answer/Confidential	UMT, Compeq, Tripod	No answer

**The Top 3 PCB Vendors Per Number Of Appearances: Tripod; Hannstar; GCE
HuaQin Is Using A Lot Of PRC PCB Vendors**

Bare PCB (3 of 3)

150°C Is The Most Common Tg



Is there a cost impact (state increase or decrease %) by using black solder mask instead of green solder mask?				
Compal KS	Compal CD	Compal CQ	HuaQin	Inventec CQ
No	N/A	NA	cost 3% up	No answer
LCFC	MSI	Pega CQ	Pega SZ (JDM1 + Intel SPG & NSG)	Wistron
No answer	No answer	cost 5% up	MSFT all black PCB	No answer
What is the Tg (Glass transition Temperature) of PCB used for Intel TigerLake TGL UP3 package?				
Compal KS	Compal CD	Compal CQ	HuaQin	Inventec CQ
Tg150	No answer	Tg170	Tg150	Tg150
LCFC	MSI	Pega CQ	Pega SZ (JDM1 + Intel SPG&NSG)	Wistron
Tg150	Tg150	Tg150	Tg150	No answer
What is the Tg (Glass transition Temperature) of PCB used for Intel TigerLake TGL UP4 package?				
Compal KS	Compal CD	Compal CQ	HuaQin	Inventec CQ
HIMB : Tg 150; RIMB: Tg > 155	No answer	N/A	Tg150	Tg150
LCFC	MSI	Pega CQ	Pega SZ (JDM1 + Intel SPG & NSG)	Wistron
Tg150	N/A	N/A	Tg150	No answer

LTS (Low Temperature Solder)

LTS (1 of 3)

Compal CD	HuaQin	Inventec CQ	LCFC	Pega CQ	Pega SZ (JDM1 + Intel SPG&NSG)	Wistron
Which OEMs have you discussed LTS Assessments?						
Yes, Dell	No	No	Lenovo	ASUS	Intel SPG (NUC)	Intel
State LTS Solder Paste model (Vendor Name, Formulation Name, Particle Mesh Size) that you are using today for HVM						
	Eunow DW-1000,Type4		Alpha/HRL1 Type4; Senju L28	Alpha/HRL1; Senju L201-N146; Eunow DW-1000	Eunow, DW-1000, Type4	ShenMao
What are the LTS paste formulation (Name, Particle mesh type) you are currently evaluating for future use?						
	Eunow DW-1000,Type4	Shenmao , Eunow & Tumura	Shenmao/汉尔信	No	Eunow, DW-1000, Type4	ShenMao
Do you have restriction on LTS paste based on Halogen level (example ROLO or ROL1)?						
	ROLO	Current no, Align with SAC, Customer require halogen free	ROL1	ROLO	No	ROLO
Do you have restriction on specific elements contained in LTS paste (example Antimony 锑 Sb, Other)?						
	No, depends on customer	No	No	No	No	No

LTS (2 of 3)

Compal CD	HuaQin	Inventec CQ	LCFC	Pega CQ	Pega SZ (JDM1 + Intel SPG&NSG)	Wistron
How long would it take to qualify a new LTS paste material before use in HVM?						
	depends on customer	At least 1 year	6-12months	3~5 months	3~6 month	Depends on customer requirement
What are the LTS paste qualification methodologies and metrics (pass/fail criteria) covering Solder Joint Quality (SJQ) and Solder Joint Reliability (SJR)?						
Solder strength, void, SJ opens, drop, pull/push test	Follow SAC305, focus on drop test	x-section, D&P, thermal shock	x-section, D&P, Drop, ORT		x-section, D&P, component push/pull test	
What is the minimum number of LTS paste options you would need to qualify to support an HVM product?						
2	2	2	2	2	2	Depends on customer requirement
What types of products (market segments) are you currently using LTS paste formulation?						
			Notebook		Intel NUC	Confidential
Have you experienced any issues w/LTS formulation for any components used?						
	Need LTS stencil recommendation for BGA		No		No	No

LTS (3 of 3)

Compal CD	HuaQin	Inventec CQ	LCFC	Pega CQ	Pega SZ (JDM1 + Intel SPG&NSG)	Wistron
Do you foresee any special controls needed (i.e. dedicated line & labeling) for LTS implementation?						
Dedicate LTS label	Dedicate line	Shopflow system	Laser Mark		Track solder paste by shopflow system	Dedicate line and tools
What is your Product Quality and Reliability Certification for a LTS System						
	Follow customer	SJ reliability	Follow SAC305			Depends on customer requirement

X-Ray FA Capability

X-Ray FA Capability (1 of 3)

Is the Detector - Camera based or Flat Panel?				
Compal KS	Compal CD	Compal CQ	HuaQin	Inventec CQ
Flat panel		Flat Panel	Camera	Flat Panel
LCFC	MSI	Pega CQ	Pega SZ (JDM1 + Intel SPG & NSG)	Wistron
Flat Panel	Matrix X3, Flat Panel	Flat Panel	Flat Panel	Flat Panel
What is the resolution (Mega Pixel) for detector used?				
Compal KS	Compal CD	Compal CQ	HuaQin	Inventec CQ
950nm		1.3Mega pixel		3um
LCFC	MSI	Pega CQ	Pega SZ (JDM1 + Intel SPG&NSG)	Wistron
	10um		0.5um	0.5~10um
Does the source have a built in Filter for X ray Dosage limitation? If Yes, what is the filter material and thickness (mm)?				
Compal KS	Compal CD	Compal CQ	HuaQin	Inventec CQ
No		No	No	No
LCFC	MSI	Pega CQ	Pega SZ (JDM1 + Intel SPG & NSG)	Wistron
No	Yes, copper, 0.4mm	No	No	Depends on customer requirement

X-Ray FA Capability (2 of 3)

What is the maximum angle (+/- degrees) for Oblique (2.5D) viewing?

Compal KS	Compal CD	Compal CQ	HuaQin	Inventec CQ
+/-70		+/- 65	45	45
LCFC	MSI	Pega CQ	Pega SZ (JDM1 + Intel SPG & NSG)	Wistron
50~60	45	45	+/-70	70

What is the minimum distance (mm) of Xray tube Source to Object (BGA device)?

Compal KS	Compal CD	Compal CQ	HuaQin	Inventec CQ
50mm		5mm		800mm
LCFC	MSI	Pega CQ	Pega SZ (JDM1 + Intel SPG&NSG)	Wistron
2~5mm	5mm	50mm	Theoretically 0mm	

Is the Source emitting X ray radiation applied from top to BGA device or is it from bottom below the PCB?

Compal KS	Compal CD	Compal CQ	HuaQin	Inventec CQ
Top		Bottom	Top	Bottom
LCFC	MSI	Pega CQ	Pega SZ (JDM1 + Intel SPG & NSG)	Wistron
Bottom	Top	Top	Bottom	We have both kind machine

X-Ray FA Capability (3 of 3)

Is there a limitation specification for X-Ray (RADS/mGray) Dosage exposure?				
Compal KS	Compal CD	Compal CQ	HuaQin	Inventec CQ
No		No	No	No
LCFC	MSI	Pega CQ	Pega SZ (JDM1 + Intel SPG & NSG)	Wistron
No	No	No	No	3~12RAD, depends on customer requirement
What is the cumulative specification for maximum exposure in RADS/mGray per scan as measured by a dosimeter?				
Compal KS	Compal CD	Compal CQ	HuaQin	Inventec CQ
N/A		N/A	N/A	N/A
LCFC	MSI	Pega CQ	Pega SZ (JDM1 + Intel SPG&NSG)	Wistron
N/A	N/A	N/A	N/A	We use dosimeter
Are Global metal shields used between X-ray source and target SOC/Memory devices to limit dosage exposure?				
Compal KS	Compal CD	Compal CQ	HuaQin	Inventec CQ
No		No	No	No
LCFC	MSI	Pega CQ	Pega SZ (JDM1 + Intel SPG & NSG)	Wistron
No	No	No	No	Our new machine can meet the requirement w/o metal shield

SD WLAN

SD WLAN (1 of 4)

What is the preferred stencil thickness?				
Compal KS	Compal CD	Compal CQ	HuaQin	Inventec CQ
0.1mm	0.1mm		0.1mm	0.1mm
LCFC	MSI	Pega CQ	Pega SZ (JDM1 + Intel SPG & NSG)	Wistron
0.08mm or 0.1mm	0.1mm	0.1mm	0.1mm	Depends on product
Stencil Fabrication used Material? (For Example: Nickel blank/sus 403/etc...) please be specific				
Compal KS	Compal CD	Compal CQ	HuaQin	Inventec CQ
FG	FG		FG	SUS403
LCFC	MSI	Pega CQ	Pega SZ (JDM1 + Intel SPG&NSG)	Wistron
SUS316	SUS304	SUS403	FG steel + Nano Coat	Stainless steel
Is there a stencil thickness stepping up or down on Mother board SD1216 location?				
Compal KS	Compal CD	Compal CQ	HuaQin	Inventec CQ
No	Possible		No	No
LCFC	MSI	Pega CQ	Pega SZ (JDM1 + Intel SPG & NSG)	Wistron
Step up	No	No	No	Depends on product

SD WLAN (2 of 4)

Screen Printing squeegee size?				
Compal KS	Compal CD	Compal CQ	HuaQin	Inventec CQ
Follow PCB	350mm		300mm	350~450mm, depends on PCB size
LCFC	MSI	Pega CQ	Pega SZ (JDM1 + Intel SPG & NSG)	Wistron
280mm	Follow PCB	350mm	Follow product (10." 12". 14")	170~620mm, depends on product
Stencil opening for WLAN SD1216				
Compal KS	Compal CD	Compal CQ	HuaQin	Inventec CQ
Follow Intel MAS	follow intel MAS then optimize		0.7X0.26	Follow intel MAS first then do optimize
LCFC	MSI	Pega CQ	Pega SZ (JDM1 + Intel SPG&NSG)	Wistron
follow MAS, external IO pins step up to 6mils	Depends on the pitch	follow intel MAS first then do optimize	All customized based on yield	Depends on product
Do you use low temp solder? What is the melting temperature?				
Compal KS	Compal CD	Compal CQ	HuaQin	Inventec CQ
No	No		No	No
LCFC	MSI	Pega CQ	Pega SZ (JDM1 + Intel SPG & NSG)	Wistron
Yes, 151C	No	No	NUC, Yes	Yes

SD WLAN (3 of 4)

Online auto clean: Print machine automatic wipe and clean
Online manual clean: Operator pull stencil off the machine and manual wipe clean
Offline auto clean: Send the stencil to stencil wash machine

Stencil cleaning frequency?				
Compal KS	Compal CD	Compal CQ	HuaQin	Inventec CQ
Every 8Pcs print automatic clean	6H		Auto clean every 4Pcs Print Manual clean every two hours	2H online manual clean, 12H offline clean
LCFC	MSI	Pega CQ	Pega SZ (JDM1 + Intel SPG & NSG)	Wistron
Every 8Pcs online auto clean, 2H online manual clean, 4H offline auto clean	3~5pcs automatic clean, 30Pcs manual clean	Automatic clean every 3~6 print	6~8 print auto clean. 12hr manual clean	
Can the factory trace the SPI data on a board that failed for an open joint to see if there is any abnormality/out of spec on the solder paste height or solder paste volume?				
Compal KS	Compal CD	Compal CQ	HuaQin	Inventec CQ
Yes	Yes		Yes	Yes
LCFC	MSI	Pega CQ	Pega SZ (JDM1 + Intel SPG&NSG)	Wistron
Yes	Yes	Yes	Yes	Yes
Any SMT carrier or reflow pallet being used during top and bottom side reflow? Any support on the WLAN SD1216 for the SMT carrier or reflow pallet?				
Compal KS	Compal CD	Compal CQ	HuaQin	Inventec CQ
For 2nd side we will use reflow pallet	For 2nd side we will use reflow pallet		Yes	Yes, we all use reflow pallet
LCFC	MSI	Pega CQ	Pega SZ (JDM1 + Intel SPG & NSG)	Wistron
Yes	Yes, on Top side only	Yes	Yes	Yes

SD WLAN (4 of 4)

Do you reused the thrown (discarded) part from SMT pick and place process?				
Compal KS	Compal CD	Compal CQ	HuaQin	Inventec CQ
Yes	Yes		Yes	Yes, follow the RFC of thrown part
LCFC	MSI	Pega CQ	Pega SZ (JDM1 + Intel SPG & NSG)	Wistron
Yes, will visually confirm no cosmetic issue, and mark on the component body	Yes	Yes	Yes	Depends on customer requirement
Do you place the WLAN SD on top (1 time reflow) or bottom (2 times reflow) side of the motherboard?				
Compal KS	Compal CD	Compal CQ	HuaQin	Inventec CQ
Both	Top side		Top Side (Current there is 1 design at bottom side)	Top side
LCFC	MSI	Pega CQ	Pega SZ (JDM1 + Intel SPG&NSG)	Wistron
Top side	Top side	Both	Top side	Depends on product



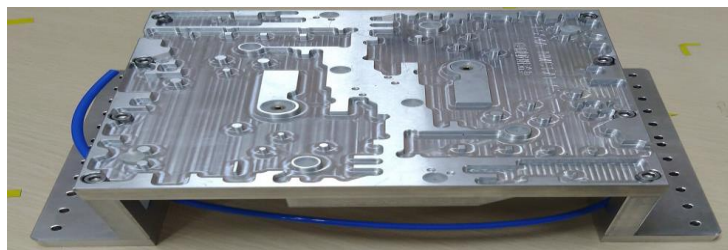
Customer Pictures Of Support Block



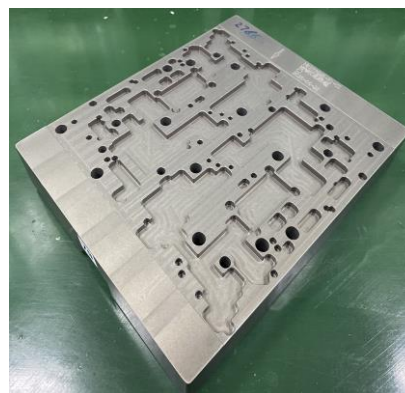
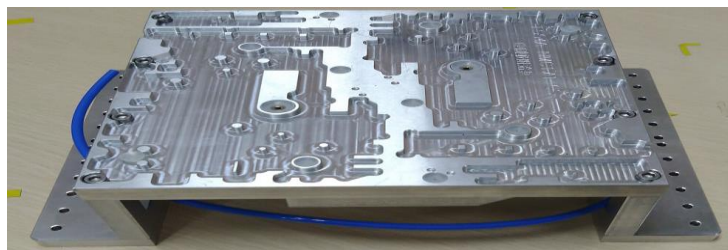
Vacuum Block



Compal KS



LCFC



HuaQin



Pega CQ

Support Block

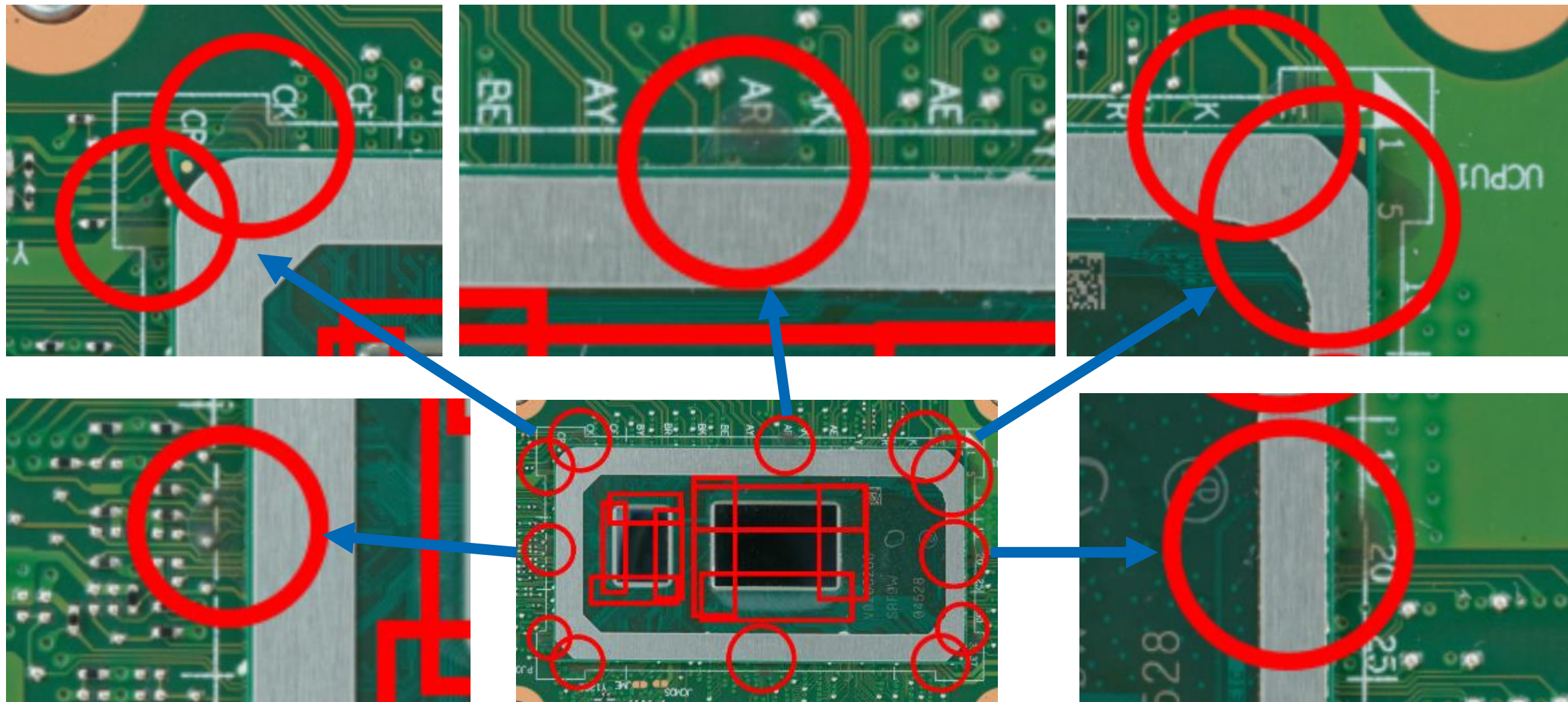


Inventec CQ



Pega SZ

Lenovo X390/LTS Tear Down Picture

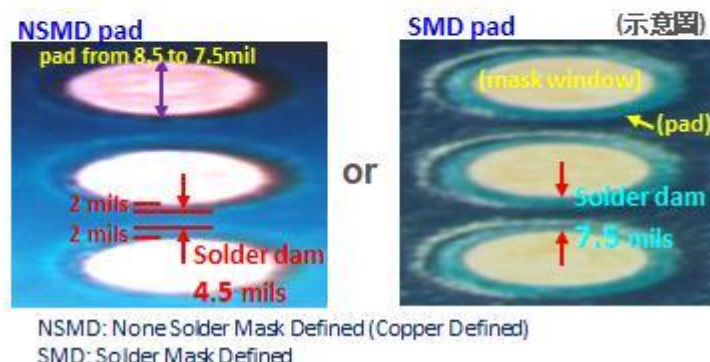


Compal's Learning About Why Black SM Not Easy To Have Good MD Pad In Small Pitch



一般顏色的防焊漆寬度 <3mil 時, 在曝光&顯影製程時, 防焊漆易被高壓的顯影藥液冲刷脫落, 而黑色油墨在曝光作業時, 其表層會先吸收光能而固化, 使得底層較易產生固化不全, 因此黑色油墨需要較寬的 solder dam (防焊壩) 以加強其與 PCB 基材的抓持力, 避免被高壓的顯影藥液冲刷脫落. 考量各家廠商製程能力不同, 目前統一的要求是防焊寬度至少要做到 4mil 以上.

以 0.4mm pitch QFP/QFN 零件, 如果使用 NSMD pad (下圖左), 則必須將 pad 寬度改小, 但可能間接削弱焊錫結合力. 如果使用 SMD pad (下圖右), 則可以維持原先的裸銅焊墊面積不變, 也可使得 solder dam=7.5mil (>4mil).



During PCB mfg process, the SM (Solder Mask) will be easier washed away by high pressure development-fluid, if the solder mask dam < 3mil.

When it comes to black solder mask, during light exposure process, the top surface of the SM will absorb more light energy thus cured quickly, lead to the SM bottom side still soft at the end of the process, weakening the capability of SM dam to hold itself on top of the PCB surface. The SM dam will be easily washed away at the following process 'high pressure development-fluid wash'.

So in our design guideline, we request at least 4mil SM dam width for black SM PCB. It limited the possibility to have MD pad in small pitch BGA because MD pad have much narrow SM dam compare to SMD pad